Environment, Development and Sustainability (2025) 27:16371–16391
https://doi.org/10.1007/s10668-024-04589-3

Impact of ethical leadership and psychological ownership on employees' voluntary pro-environmental behaviour: a case study of selected organisations in Delta State Nigeria

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Received: 24 April 2022 / Accepted: 30 January 2024 / Published online: 25 February 2024
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Abstract
There is a need to complement strategic policies with discretionary environmentally friendly behaviour to ensure sustainable performance is recognised and accepted. This research investigates the impact of ethical leadership and psychological ownership on employees' voluntary pro-environmental behaviour and whether psychological ownership amplifies the impact of ethical leadership on employees' voluntary pro-environmental behaviour. The study was conducted amongst employees of selected public and commercial organisations in Nigeria. A cross-sectional survey of 163 employees provided the analysed data. Data were collected using a self-administered questionnaire, including three established scales. The mean age of the participants was 36.34, with a 7.88 standard deviation; men were 43.6%. Data analysis was conducted using statistical regression complemented with Process Macro. The findings indicated that ethical leadership has a positive association with voluntary pro-environmental behaviour ($\beta = 0.64$, 95% CI [0.46–0.68], $t = 10.53$, $p < 0.001$). Furthermore, psychological ownership was also found to predict voluntary pro-environmental behaviour as well ($\beta = 0.23$, 95% CI [0.10–0.35], $t = 3.56$, $p < 0.001$). However, psychological ownership did not moderate the relationship between ethical leadership and voluntary pro-environmental behaviour. Since this is a pioneer study, it could be tentatively concluded that although ethical leadership and psychological ownership energised the voluntary pro-environmental behaviour of the employees, their interaction did not.

Keywords Ethical leadership · Psychological ownership · Pro-environmental behaviour · Sustainability · Social learning theory

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1 Introduction

Environmental sustainability is now a core concern for society, considering the catastrophes engulfing the physical environment; a call epitomises sustainable development goals for concerted efforts to preserve the ecosystem. The intensity of the degradation calls for evaluation and re-evaluation of policies and behaviour related to the environment within several spheres of human involvement, such as home and organisation. Given organisations' crucial impact on the environment's undesirable state (Fatoki, 2021; Saleem et al., 2021), they are under high pressure to reduce the damage (Sulphey et al., 2024). However, there is a history of research productivity on organisational sustainability practises, but it is still in its infancy, as it has yet to cover several dimensions sufficiently. In this category of sustainable workplace practises, minimal research attention is paid to employee-initiated behaviour with implications for the environment.

Consequently, this research focussed on employees' voluntary pro-environmental behaviour (VPEB) or environmental behaviours that organisations do not compel, reward, or officially acknowledge (Kim et al., 2023; Sulphey et al., 2024). The positive outcomes of employees' voluntary and less apparent behaviour extend to individuals, organisations, and the pursuit of environmental sustainability (Shah et al., 2021). Several phases, such as "organisational citizenship behaviour for the environment," "organisational environmental citizenship behaviour," and "organisational citizenship behaviour directed towards the environment," have been used to illustrate the employees' voluntary care for the environment. Other terms include "green organisational citizenship behaviour", "voluntary workplace green behaviour", and VPEB of employe es (Abouelenien et al., 2024; Fatoki, 2021; Ullah et al., 2021).

Despite the official and usual top–down initiatives that affect the pro-environmental behaviour of employees, it is well documented that some employees voluntarily engage in sustainability. Therefore, it is necessary to appreciate the mechanisms related to environmental sustainability issues in the workplace, as individual sustainable practises contribute to organisational sustainability (Cheng et al., 2021; Fatoki, 2021). This understanding of the role of individual sustainability in organisational success is essentially represented in the definition of VPEB as personal discretionary actions that support formal environmental policies and improve environmental performance (Fatima & Azhar, 2021). VPEB includes employee-initiated knowledge sharing on pollution prevention, solutions for waste reduction, recycling, and reuse of work materials (Sulphey et al., 2024).

Furthermore, the formal control mechanisms such as environmental safety systems and monitoring operations that characterised organisations have not effectively managed environmental challenges (Fatima & Azhar, 2021; Fatoki, 2021). Therefore, it becomes imperative to have a deeper and better understanding of VPEB by pinpointing precursors to offer insights to human resource professionals on how to encourage the behaviour (Cheng et al., 2021). Several researchers (e.g. Saleem et al., 2021) have noted the need for further studies on identifying plausible predictors of employees' VPEB. Similarly, some reviews (see, e.g. Blankenberg & Alhusen, 2019) have proposed studying the determinants of VPEB, from examining individual variables to investigating the interplay of factors. However, a survey of the extant literature reveals that the observations and concerns raised by these researchers and others have not been extensively studied through empirical research.

The typical recommendation in the literature is the need for research on the influence of leadership types and their interaction with other variables on the voluntary environmentally friendly behaviour of employees (Blankenberg & Alhusen, 2019; Khan et al.,

2021; Ullah et al., 2021). Despite advancements in research on the correlation between leadership styles and VPEB, there is a noticeable gap in the literature studying the relationship between ethical leadership (EL) and VPEB (Cheng et al., 2021; Yusop & Adam, 2021). Put differently, there is a scant study on the relationship between EL and VPEB of the employee. Ethical leadership refers to the normative display of conduct expressed in personal actions and relationships with others and encouraging such conduct in followers through shared communication, decision-making, and reinforcement (Edevbie & McWilliams, 2023). Studies of EL and VPEB employees are necessary for contemporary workplaces saddled with unethical conduct (Don-Solomon & Ayawei, 2021; Mitchell et al., 2020).

The role of psychological ownership (PO) in moderating the connection between EL and VPEB is not well understood, as related research is scant. However, considerable research has been conducted, and sufficient confirmation has been obtained on the potential of PO to mediate the influence of various leadership styles on employee VPEB (Cheng et al., 2021; Ullah et al., 2021; Waqas et al., 2021). The valuable insights emerging from these studies that search for mediators need to be complemented by moderating studies to understand effective leadership styles in the VPEB of employees comprehensively.

The previous presentation established gaps in leadership, PO and employee VPEB literature. This gap can be found in the scant study on the relationship between EL and employee VPEB and the lack of study on whether PO is a moderator in the relationship between EL and employee VPEB. This study sought to address these gaps by examining the associations between these variables. The different issues addressed in this investigation were categorised into subsequent research inquiries. Do EL and PO positively predict employee VPEB? Does PO moderate the relationship between EL and the VPEB of employees? One of the three goals is to evaluate how EL affects the VPEB of employees. Second, it evaluates the impact of PO on the employee's VPEB. Third, it clarifies whether PO influences how the EL and VPEB of the employee interact amongst workers in service firms. whilst the relationships and interactions of the focal variables are examined, the overriding purpose of this study is to identify pliable fosters of employee VPEB.

This study was conducted within certain limits. First, whilst public and private organisations were represented in the sample, manufacturing organisations were not, as all participants were drawn from service organisations. It indicates that the findings of this study may not fully fit manufacturing organisations since some differences in the behaviour of manufacturing and service organisations' employees have been expressed (Ahmad et al., 2021; Wall, 2021). Second, the identification of leadership styles is substantial and inclusive. The recent list of leadership styles in the literature includes ethical, spiritual, transactional, transformational, supportive, authentic, servant, and positive (Rothmann & Cooper, 2022). These various forms of leadership can influence the VPEB of the employee. However, this research was constrained to EL, and as certain leadership styles may not show substantial correlation, the study results may not fully represent all leadership styles.

The structure of the study begins with the conceptual background presented in Sects. 1 and 2, which comprises a review of critical variables and research hypotheses. Section 3 elucidates the methodology. This section encompassed details on the research design, sample, measures, and statistical tests used in the study. It also included discussions on variance and control variables. Section 4 presents the results related to the reliability and validity of the adopted scales and the tests conducted on the hypotheses. Section 5 discusses the results and the implications for theory and practise. The article ends with Sect. 6, which offers conclusions, limitations, and recommendations for future research.

2 Theoretical background and research hypotheses

The thesis of this study is that EL and PO promote employee VPEB with a rider that Po has a moderate influence on the relationship between EL and employee VPEB. Social learning theory (Bandura, 1977) and social identity (Tajfel & Turner, 1979) formed the basis for the postulations. Social learning theory explains the proposed relationship between EL and employee VPEB, whilst social identity accounts for the relationship between PO and employee VPEB and the moderation of PO in the relationship between EL and employee VPEB. Social learning theory proposes that learning occurs through observation, imitation and modelling and that factors such as attention, motivation, attitudes, and emotions influence the process. The principle of social learning theory is that people observe others' behaviour and attitudes and the consequences of the attitude and behaviour and then use the information to guide their actions. The theory recognised the interaction of environment and cognition in how people learn. Ethical leaders are characterised by altruism, caring for sustainability, morality, compassion, and fairness (Kim et al., 2023; Risbjerg Nørgaard, 2022), which could reflect their perception of the environment. In addition, concern for the environment largely represents these elements. Therefore, in their disposition, ethical leaders would be concerned with the environment, which could lead to related action. According to social learning theory, the employees could imbibe these qualities and actions of the ethical leader. In other words, employees would learn VPEB from their ethical leaders.

Social identity refers to an individual's understanding of belongingness to specific social groups and the emotions and values associated with group membership (Tajfel & Turner, 1979). Social identity theory proposes that people strive to achieve positive self-esteem and that group membership can enhance or lower someone's self-esteem and self-concept. An organisation is a group of people who come together for a specific purpose. PO represents a strong identity within the organisation. Employee VPEB is discretionary employee behaviour with positive implications for the organisation. According to social identity theory, since the rating of an organisation has implications for employee self-esteem and that VPEB is discretionary behaviour of the employee that can enhance the rating of the organisation, the level of an employee's feeling of belongingness, identity, and ownership will relate with the performance of VPED.

Similarly, social identity theory also accounts for the proposed moderation role of PO in the relationship between EL and employee VPEB. For social identity theory, although EL has potential characteristics for VPEB of employees, the expression of the behaviour will sufficiently depend on how much employees identify with the organisation. Some studies (see, e.g. Wang et al., 2022) have established the validity of studying the relationship between PO and employee environment-friendly behaviour.

2.1 EL and VPEB of the employee

Ethics leadership is the demonstration of socially acceptable behaviour through one's behaviour, interactions with others, and disclosure of such behaviour or attitude to followers (Kim et al., 2023). Several leadership styles, such as responsible, authentic, and servant leadership, have been identified and examined in the literature on organisational behaviour. However, in the present times of gross moral decadence, EL is gaining much research interest, as it is strongly linked to followers' ethical conduct and attitudes. Ethical leaders model appropriate behaviour through decision-making, reinforcement, and two-way

communication. An ethical leader works to uphold ethical standards and exercise moral leadership. Justice, power sharing, integrity, people-orientedness, and ethical direction are amongst the essential traits of EL (Saleem et al., 2021).

In general, pro-environmental behaviour is individual behaviour that reduces negative environmental impact. Such behaviour can be employee VPEB or involuntary (work-task-related ecologically friendly behaviour). Although formal is discretionary, organisations officially mandate the latter (Alzaidi & Iyanna, 2022; Fatoki, 2021). Employee VPEB is a discretionary act that organisations do not officially recognise or reward, but it has implications for environmental sustainability (Khan et al., 2021). It shows how much employees go above and beyond their job requirements to act in environmentally friendly behaviour. Such behaviour could be a waste reduction at work, recycling, conserving energy, reducing paper usage, proposing work suggestions in environmentally friendly ways, and encouraging colleagues to promote pro-environmental behaviour (Sulphey et al., 2024; Ullah et al., 2021).

EL is postulated to favour the employee's VPEB based on the linked qualities, the existing empirical literature, and the social learning theory (Bandura, 1977). EL symbolises justice, fairness, equity, morality, ethical climate, social responsiveness, responsibility, and accountability. Several characteristics have been identified as precursors of VPEB (e.g. Misch et al., 2021). Similarly, the results of substantially related studies positively implicate leadership styles in VPEB. For example, responsible, reliable, and compassionate leadership styles positively impact employee VPEB (eSahar et al., 2023; Ullah et al., 2021). Moreover, social learning theory proposes that new behaviour is learned by mimicking and observing others (Saleem et al., 2021; Yen, 2022). According to social learning theory, an ethical leader can influence followers to act ethically through communication, modelling, rewards, and punishment. Several studies (e.g. Sarwar et al., 2023; Yen, 2022) confirmed the social learning theory in organisational settings. Accordingly, it was hypothesised that.

Hypothesis 1 Ethics leadership positively predicts employee VPEB.

2.2 PO and employee VPEB

PO consists of cognitive and affective components that fulfil essential human motivations and needs, which include belongingness and self-identity (Nwanzu & Babalola, 2022a). It is a mental state where individuals develop a sense of ownership towardss a particular target (Ullah et al., 2021). It reflects the belief that individuals own a target, wholly or partially (Tanyildizi & Doğan, 2023; Van Dyne & Pierce, 2004). PO is a feeling of personal claim to something, 'Mine' (Peck et al., 2021). Thus, PO indicates a scenario where employees perceive the organisation as belonging to them (Cheng et al., 2021; Yusop & Adam, 2021). This ownership emotion could make employees perform better, develop self-efficacy, and feel more responsible to the organisation. Based on relevant attributes, prior empirical research, and insights from social identity theory (Tajfel & Turner, 1979), this study posited a positive influence of PO on employees' VPEB. PO encompasses various aspects of the organisation, including pride, identification, and commitment. These have been positively linked with several forms of voluntary ecologically friendly behaviour of employees (see, e.g. Shah et al., 2021).

Empirically, PO positively correlates with an employee's VPEB (Jiang et al., 2019). PO positively predicts green knowledge sharing, creativity, corporate green operations,

green organisational behaviour, and product development (Chang & Hung, 2021; Ozbozkurt et al., 2022; Suessenbach et al., 2018; Wang et al., 2022). Theoretically, social identity pertains to an individual's understanding of belongingness to specific social groups and the emotions and values associated with group membership (Tajfel & Turner, 1979). Therefore, when employees identify as part of an organisation, they are more inclined to engage in positive behaviours. Employees are expected to judge their success in terms of the success of the organisation of which they are members. Therefore, the thesis is that employees identifying with their employer will express positive work behaviour, such as VPEB. Consequently, this study hypothesised that.

Hypothesis 2 PO positively predicts employee VPEB.

2.3 PO as a moderator in the EL and VPEB relationship

This study also investigated PO as a moderator in the relationship between EL and VPEB. The existing networks of intervening relationships amongst these variables informed the moderation proposition of this study. For example, PO mediated between responsible, reliable, compassionate leadership styles and VPEB of employees (Ullah et al., 2021). PO served as a mediator between organisational citizenship behaviour towardss the environment and empowerment of leadership (Jiang et al., 2019) and the relationship between practises that enhanced employee participation, green motivation, and green behaviour (Waqas et al., 2021). Furthermore, EL was found to mediate the link between corporate social responsibility and VPEB partially, moderate the interaction between green intentions and green behaviour, and regulate the association between green human resource practises and VPEB (Ahmad et al., 2021; Wu et al., 2021). The central thesis is that both EL and PO need to be appropriately combined to maximise their positive impact on the VPEB of employees. It is plausible that employees with higher PO would identify more with the organisation and be more likely to express more positive organisational behaviour, such as VPEB. Therefore, this study hypothesised that.

Hypothesis 3 The impact of EL on the VPEB will be moderated by PO, making the link more significant when PO is high than when it is low.

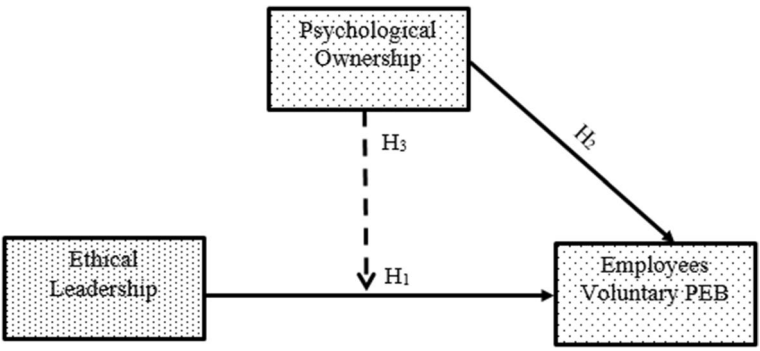


Fig. 1 Research model

Figure 1 illustrates the conceptual framework of the present study, depicting the interrelationships between VPEB, EL, and PO. The model is causally recursive, as feedback effects are not expressed (Nayebi, 2020), whilst EL and PO influence the VPEB of employees, not otherwise. Furthermore, the framework presents the PO as a moderator in the EL and VPEB relationship.

3 Methodology

3.1 Participants

The study used a quantitative and cross-sectional research design, examining relationships between latent constructs (Zaman et al., 2021). The sample comprised 163 employees of various public organisations (64%) and private organisations (36%) in Delta State, Nigeria. In this research location, most formal employment is in government-owned organisations. The average age ranged from 23 to 57 years ($\bar{x}=36.34$; SD = 7.88). Most of the participants were women (56.4%) and had education levels below a first degree (33%), a first degree or its equivalents (60%), and a post-graduate certificate (7%). Participants were literate, and these characteristics conferred validity to the adopted self-report measures. The sample comprised every category of employees, which enhanced the ecological validity. The respondents were in permanent employment and covered junior and senior staff members, with diversified samples from the organisation's two main classifications (public and private). This study incorporates participants from both public and private organisations because conventional theories that have suggested substantial differences in the leadership of these two types of organisations are now being challenged by organisational behaviourists (Flemming, 2016).

More recently, Seitkazin (2020) concluded that the answer to whether leadership styles are distinctive in the private and public sectors might be "yes" or "no". Several aspects of EL are transferable between the private and public sectors (Heres & Lasthuizen, 2012). In addition, specific studies that observed leadership differences between public and private organisations never established the significance of the difference. For this study, a sample of public and private sector employees has implications for the broader application of the findings (Asim et al., 2021). The sample size for this study follows the heuristic principle, as it adopted the general rule provided in the literature (Lakens, 2022). For example, the sample size met the recommendation of a minimum of five participants for each item measured in a study or a minimum of 20 participants for every focal variable of a study (Rahman, 2023; Zaman et al., 2021). Hence, this study sample size was deemed sufficient, considering the investigation of three focal variables totalling 27 items. Additionally, the chosen sample size adhered to the principle that when the anticipated effect size is uncertain, a sample size suitable for a medium effect size is satisfactory; as such, a sample size of 163 provides more than 80% power to detect a relationship at a significance level of 0.05 if it exists (Althubaiti, 2023).

3.2 Variables

This study involved three focal variables: independent, dependent, and moderator and three control variables: sex, age, and educational qualification.

3.2.1 Independent variables

This study conceptualised and analysed EL and PO as independent variables. Several studies have associated these variables in the existing literature as predictors of valuable employee outcomes, including extra-role behaviour, job satisfaction, knowledge sharing, and work engagement (Junaidi, 2023; Phetsombat & Na-Nan, 2023; Su et al., 2021). These variables were identified to play a third role in several relationships between variables in an organisational setting. The importance of such a relationship pattern is that it could serve as a basis for causal inferences.

3.2.2 The moderator variable

PO was also examined as a moderator variable in totalling the independent variables. The existing literature showed that substantial studies had examined PO as a mediator in the link between EL and the VPEB of employees (Cheng et al., 2021; Ullah et al., 2021). Nevertheless, only a few studies have explored PO's role as a moderator in the interactions between EL and VPEB. The emerging findings from studies where PO has acted as a mediator have been valuable for theoretical and practical purposes. This recognition emphasises the importance of investigating the moderating role of PO in the relationship between EL and VPEB of employees.

3.2.3 The dependent variable

The VPEB of the employees served as the study's dependent variable. The importance of sustainable development cannot be overstated for the individual, the organisation, or society. Many people consider sustainable development a higher-order construct with three aspects: economic, social, and environmental (Terra dos Santos et al., 2023). Although the dependent variable has implications for economic and social aspects, it primarily focuses on the environmental aspect of sustainable development. A global concern for environmental sustainability catalyses research to understand the process that could lead to achievement. Within that concern, this study examined leadership and attitudes that could lead to employee behaviour that positively impacts environmental sustainability.

4 Research design and statistics

The cross-sectional study design used a practical convenience sampling technique to select participants who received the questionnaires. The chosen sampling technique is suitable for this study as it enables researchers to select participants based on their judgement and preferences aligned with the study objectives. According to Zaman et al. (2021), a cross-sectional design is appropriate because the study aimed to establish associations between latent constructs. It also enables the simultaneous assessment of several variables (Howitt & Cramer, 2017). The study improved the validity of the results obtained through the cross-sectional design using the procedural approach to control common-method variance and analyse control variables (Spector, 2019). Regression

analysis and the PROCESS macro for SPSS were used to test the study hypotheses. This statistical technique was adequate, as this study predicted the VPEB of the EL and PO employees.

Furthermore, the PROCESS macro has attracted much recognition and acceptance amongst researchers and has been presented as a promising approach to assessing moderation (Hayes, 2012).) As regression is a parametric statistical test, the design and preliminary data analysis considered several assumptions (Babalola & Nwanzu, 2022), such as independent data, interval scaling, and linearity associated with its appropriate usage. For example, the data collected from the participants were independent of each other, meeting the different response requirements. The items of the adopted measures were prepared along the Likert scaling format anchored on a 5-point scale. The Likert scaling format is accepted in the literature to satisfy the demand for the measurement interval level (Matthews, 2017; Wu & Leung, 2017). The scatter plots in the data sets showed a linear association between the focal variables, which satisfies the linearity assumption. The range of zero-order correlations reported in Table 1 indicated that the data set for this study met the autocorrelation assumption.

Furthermore, the distribution characteristics with skewness and kurtosis were tested. The skewness for PO was −0.28, EL, −0.58, and VPEB of the employee, − 0.37 with kurtosis, −0.36, −0.22, and −0.93 for PO, EL and VPEB, respectively. These statistics fall within the acceptable level of normal distribution (Korkmaz, 2020; Madsen, 2011).

4.1 Procedure

Data for this study were collected with self-reported measures. Two hundred questionnaires were distributed to workplace participants; 171 filled questionnaires were received in approximately one month. Physical checks revealed that eight participants did not properly treat their questionnaires. Therefore, analyses were performed on responses from 163 participants and following the low and declining trend of the response rate for the survey (Luiten et al., 2020), the response rate of 85.5% and the usable rate of 95.3% obtained in this study were adjudged satisfactorily. Individual data were collected, and mean scores were used to analyse the data. The mean score for each measure was calculated by dividing the total score by the total number of valid responses, effectively mitigating the influence of potential outliers on the data sets.

Table 1 Mean, standard deviation, alpha, and intercorrelation of study variables

	Variable	$\bar{x}$	SD	No. of item	α	1	2
1	Ethical Leadership	3.66	0.61	10	0.88	1	
2	Psychological ownership	3.39	0.64	7	0.82	0.19*	1
3	Employee's voluntary PEB	3.46	0.55	10	0.77	0.64**	0.27**

Employee's voluntary PEB = Employee's voluntary pro-environmental behaviour;

**$p < 0.001$; * $p < 0.05$

4.2 Measurement instruments

4.2.1 EL measure

Brown et al.'s (2005) 10-item scale was utilised to measure EL. The scale was based on social learning theory and assessed employees' perceptions of their direct manager/supervisor regarding fairness, trust, and ethical behaviour. Responses to the scale items were collected using a 5-point Likert-type scale, ranging from 1—Strongly disagree to 5—Strongly agree. Previous studies by Li et al. (2021) and Saleem et al. (2021) have reported satisfactory reliability and validity indices for this scale.

4.2.2 Measurement of PO

PO was measured using the one-dimensional, 7-item scale developed by Van Dyne and Pierce (2004). Participants rated their agreement with the scale items on a 5-point Likert-type scale, ranging from 1—Strongly disagree to 5—Strongly agree. The psychometric properties of the PO scale were reported to be adequate by its creators and other researchers or users, including Asim et al. (2021) and Nwanzu and Babalola (2022a).

4.2.3 VPEB measures

Robertson and Barling's (2017) 10-item organisational citizenship behaviour for the environment scale was utilised to assess employees' VPEB. Participants rated their responses on a 5-point Likert scale ranging from 1 (never) to 5 (always). The scale's authors and users (e.g. Fatoki, 2021) reported satisfactory psychometric properties. For all variables, analyses were performed at the composite level. Data analysis was performed on mean scores. The mean score for each scale was calculated by dividing the combined scores by the total number of valid responses, resulting in scores ranging from 1 to 5. The five-point Likert scale format was adopted for all variables to generate enough response variability, giving validity to statistical output. A sample of the questionnaire used in this study is shown in the Appendix.

4.3 Common-method variance

In this study, cross-sectional research design and self-reported questionnaires were adopted for data collection, and these methodologies are extensively reported to common-method bias (Bozionelos & Simmering, 2022). The effect of common-method variance (CMV) reveals inflated or deflated associations amongst measures (Liang et al., 2021). This study implemented procedural control and statistical evaluation to manage the common-method variance level. The control procedures were achieved through the study design. For example, whilst EL and PO were anchored on "agreement," the employees' VPEB measures were on "frequency." The study variables were administered to the participants on separate sheets of paper. Each measure contained a moderate and adequate number of items. Participants were assured of the voluntary nature of their participation, anonymity, and confidentiality (Nwanzu & Babalola, 2022b; Okun et al., 2020). The data sets undergo the Harman one-factor procedure, a diagnostic technique used to identify whether common-method variance could potentially be a limitation in the study. The test revealed that factors with an eigenvalue explained more than 74% of the total variation, whilst the first factor explained

27%. Validity and interpretation of the study findings are not significantly affected by common-method variance because the variation explained by the first factor is less than 50% of the total variation (Saleem et al., 2021; Waqas et al., 2021).

4.4 Control variables

The variables of age (in years), sex (men = 0, women = 1), and educational qualification (in years) were built-in as covariates in the data analysis. Several related studies (e.g. Cheng et al., 2021; Li et al., 2021; Saleem et al., 2021) examined these demographics as control variables. More so, theories such as social role theory (Eagly & Wood, 2012), socio-emotional selectivity theory (Khetjenkarn & Agmapisarn, 2020) and human capital theory (Marginson, 2019) implicate the demographics in the relationship amongst this study's focal variables. Social role theory accounts for sex differences, socio-emotional selectivity theory acknowledges age differences, and the human capital theory explains the influence of educational qualifications on certain employee work behaviours.

5 Results

5.1 Descriptive analysis

The means statistics show that the scores for the variables are relatively high. Table 1 presents the descriptive statistics for the research focal variables EL, PO and VPEB, with observed mean scores of 3.66, 3.39, and 3.46, respectively. The mean scores were above average since each was obtained from a 5-point scale. The intercorrelation statistics of the variables were positive and significant, with 0.19, 0.27, and 0.64. The level of the correlation coefficients indicates that the data analysis was not biassed by multicollinearity. The Durbin–Watson test statistics of 1.65 and 1.98 indicate that the level of autocorrelation is within an acceptable range.

5.1.1 Reliability and validity of the measures

This study used established scales (Hair et al., 2020), and their reliability and validity were also evaluated. The Cronbach alpha procedure was used to test internal consistency, and the coefficient alphas are presented in Table 1. The validity of the measures was supported by their acceptance in the literature (Karimi et al., 2020). Convergent and discriminant validity was evaluated to test the construct's validity. This study's satisfactory Cronbach alpha statistics supported convergent validity (Chikazhe et al., 2021; Zaman et al., 2021). Discriminant validity was assessed through exploratory factor analysis, which revealed that the items had higher loadings on their respective constructs than cross-loadings. This pattern of item loadings indicates discriminant validity, confirming that the items were explicitly associated with their intended factors (Chikazhe et al., 2021).

5.2 The main effect

Table 3 presents a simple regression analysis that predicts the VPEB of the EL and PO. The data in Table 2 provide evidence supporting Hypotheses 1 and 2. Regarding

Table 2 Simple regression predicting employee's voluntary PEB from EL and PO

Variables	β	SE	B	T	p	95% CI	
						LB	UB
Constant	1.35	0.20		6.70	0.001**	0.95	1.75
EL	0.57	0.05	0.64	10.53	0.001**	0.46	0.68
	$F(1; 160) = 111.06$, $p < 0.001$, $R^2 = 0.41$, Adjusted R^2 0.40, DW = 1.98						
Constant	2.67	0.22		11.94	0.001**	2.23	3.11
PO	.23	0.06	0.27	3.56	0.001**	0.10	0.36
	$F(1; 161) = 12.71$, $p < 0.001$, $R^2 = 0.07$, Adjusted R^2 0.06, DW = 1.65						

EL ethical leadership, *PO* psychological ownership; * p-value < 0.001

Table 3 Moderation of PO in the effect of EL on the employee's voluntary PEB

Variables	B	SE	t	p	95% CI	
					LB	UB
Constant	4.11	0.23	18.24	0.01**	3.66	4.55
Ethics Leadership	0.49	0.06	8.40	0.01**	0.37	0.60
Psychological Ownership	0.16	0.05	2.96	0.01**	0.05	0.27
EL * PO	− 0.11	0.07	− 1.56	0.12	− 0.24	0.03
Sex	− 0.09	0.07	− 1.31	0.19	− 0.21	0.04
Age	0.01	0.00	− 2.37	0.02*	− 0.02	0.00
Educational Qualifications	− 0.02	0.03	− 0.49	0.62	− 0.08	0.05
	$R = 0.69$, $R^2 = 0.48$, F (1; 139) = 21.18, $p < 0.001$***					

EL ethical leadership, *PO* psychological ownership, * p-value < 0.05; ** p-value < 0.01; *** p-value < 0.05

Hypothesis 1 (third, fourth, and fifth rows), EL strongly predicted VPEB ($\beta = 0.64$, 95% CI [0.46, 0.68], $t = 10.53$, $p < 0.001$). According to statistical indices, an increase in EL of one unit causes an increase in VPEB of the employee of 0.57 units. R^2 reveals a significant effect, with EL accounting for almost 41% of the VPEB variation. Variance indices demonstrate that EL can predict VPEB because the regression was statistically significant. The slight disparity between R^2 and adjusted R^2 suggests strong cross-validation, indicating the generalisability of this model to other samples within the same population. As for Hypothesis 2 (Table 2, sixth, seventh and eighth rows), PO significantly influenced VPEB ($\beta = 0.23$, 95% CI [0.10–0.35], $t = 3.56$, $p < 0.001$). An observed value of b indicates that a one-unit increase in PO results in a corresponding increase of 0.23 units in the VPEB of the employees. R^2 statistics indicate that PO accounts for 7% of the variance in predicting employees' VPEB, suggesting a small effect size. The ANOVA test demonstrates a statistically significant regression model, meaning that the VPEB can be predicted based on PO. The marginal difference between R^2 and adjusted

R^2 signifies robust cross-validation, indicating the applicability to potential samples drawn from a similar population.

5.3 Test of the moderation effect

The statistics for Hypothesis 3 are shown in Table 3. The results show similar statistics for both the simple and main effects. Simple effect analysis examined the VPEB under the condition of holding the EL or PO constant for each other. When PO was kept constant, the relationship between EL and VPEB was significant ($\beta = 0.49$, $p < 0.01$). Similarly, when EL was kept constant, the relationship between PO and VPEB was also significant ($\beta = 0.16$, $p < 0.01$). However, the combined impact of EL and PO on the VPEB did not yield significant results, leading to the conclusion that Hypothesis 3 was not corroborated. Similar results were observed when covariates were not introduced into the analysis.

Figure 2, a summary of the results, illustrates that both EL and PO have a positive and statistically significant influence ($p < 0.001$) on employees' voluntary PEB. Furthermore, it shows that PO does not moderate EL and the voluntary PEB relationship.

6 Discussion

In this investigation, the primary focus was to investigate the separate influences of EL and PO on VPEB and explore whether PO acts as a moderator in the relationship between EL and VPEB. As a result, three hypotheses were formulated and subsequently examined. Hypotheses 1 and 2 focussed on the main effects, whilst Hypotheses 3 focussed on the moderation effect. This section presents a comprehensive discussion of the findings obtained.

6.1 Main effects of EL on the VPEB of employees.

Hypothesis 1 of this study tested whether EL positively influences employee VPEB. The results observed from the simple regression analysis supported the direction of the prediction. That is, EL positively impacts the VPEB of the employees. The finding paralleled some existing related studies (Ahmad et al., 2021; Ullah et al., 2021) and aligned with

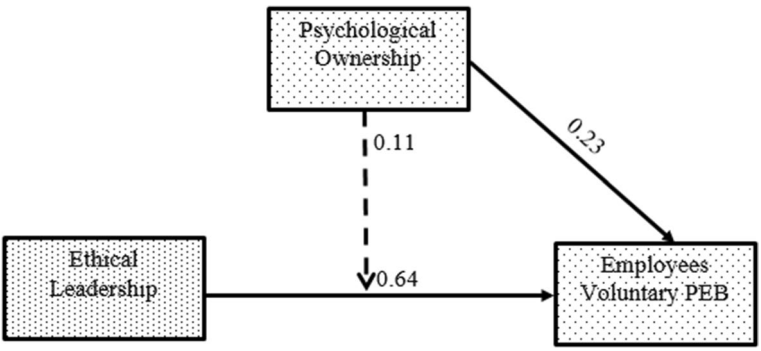


Fig. 2 Summary of findings

the associated elements of social learning theory. Ahmad et al. (2021) discovered that EL moderates the relationship between green management behaviour and green creativity and between green management practises and VPEB. Likewise, the research revealed a positive association between EL and VPEB. The observation that EL positively influences the VPEB of the employee is expected, as EL is characterised by fairness, equity, and compassion that can trigger positive behaviours, attitudes, and emotions. Several empirical investigations have confirmed the usefulness of social learning in explaining organisational leadership behaviour. For example, Byun and Lee (2021) reported that empowering high-level leaders improves the empowerment of low-level leaders.

6.2 Main effects of PO on the VPEB of employees.

Hypothesis 2 of this study tested whether PO positively influences the VPEB of employees. The result observed from the simple regression analysis supported this hypothesis. That is, PO positively influences the VPEB of the employee. This observation corresponded to that of Suessenbach et al. (2018). They reported that PO promotes VPEB. Jiang et al.'s (2019) research demonstrated a positive link between EL and employees' organisational citizenship behaviour for the environment, which was mediated by the employees' perception of PO. The result also confirmed related aspects of social identity theory and explained that PO is associated with identification and commitment. All these latter variables have the innate potential to trigger positive behaviour, such as the VPEB of employees. Several studies have implicated social identity in green employee behaviour. For example, Zhu et al. (2021) found that green organisational identity acts as a mediator, facilitating the positive impact of green management behaviour on green behaviour. Van Der Werff et al. (2021) also identified self-identity with the environment as a factor associated with various pro-environmental behaviours. Moreover, there is evidence of the moderating role of PO in connecting EL with VPEB.

6.3 Testing PO moderating roles

Hypothesis 3 tested whether PO moderates the positive impact of EL on the employee's VPEB. Data analysis indicated that the positive impact of EL on employee VPEB remained unaffected by the presence of PO as a moderator. Therefore, Hypothesis 3 was not confirmed. This finding contradicts the prediction and outcome of research like that of Ullah et al. (2021). They observed that PO mediates the association between organisational citizenship and leadership behaviour for the environment. Also, Waqas et al. (2021) found that the sense of owing to nature mediated the relationship between green management practises and employee behaviour. EL and PO predict the VPEB of the employees when either is kept constant. Furthermore, its interaction effect is insignificant, which means that the respective effects of EL and PO on the VPEB of the employees are, in practical terms, independent of each other. It means that no matter how high or low the employee perceived EL, it does not alter the effect of PO on an employee's VPEB.

In contrast, no matter how high or low the employees are in PO, it does not alter the effect of EL on the VPEB of the employees. The significant influence of EL on the VPEB could explain why PO does not moderate the relationship between EL and VPEB. The lack of moderation by PO might be attributed to EL's only substantial impact on employees' VPEB. It could be argued that since EL explains a substantial unit in the employee's VPEB, it is probably at its maximum effect on the employee's VPEB. Since EL is already

at its maximum, no level of PO could significantly alter its influence on the VPEB of the employees. It could also be argued that since the effect of PO on the employee VPEB was too small, it could not alter the enormous influence of EL on the employee VPEB. This rationale is based on Tabachnick and Fidell's (2019) observation, which states that the significance of an independent variable in a model relies on the presence and influence of other independent variables within the same set.

6.4 Theoretical contributions

This study added new theoretical insights to the literature on EL, PO and VPEB due to a lack of research on EL and VPEB; therefore, the current study focuses mainly on the relationship with VPEB. Second, this study addressed the lack of understanding about the relationship between PO and organisational citizenship behaviour for the environment. Work attitudes in the literature, such as organisational support, employee commitment, psychological green climate, and organisational identification (see, e.g. Yusop & Adam, 2021), have received substantial research and VPEB. Few studies have been conducted on the influence of PO on employee VPEB. It is a crucial research lacuna, as PO has characteristics such as identification and commitment predisposing it to several forms of VPEB.

Third, although the moderation hypothesis tested in this study was not supported, the observed results open a new research direction. It is crucial and necessary, as this study is a pioneer in investigating PO as a moderator in the effects of leadership styles on employees' VPEB. The expected research direction has potential for relevance, as studies of Ullah et al. (2021) and Waqas et al. (2021) on PO as a mediating variable in the relationship between leadership styles and employees' VPEB have produced useful, practical insights. It is worth noting that the four scales used in these studies were developed in settings different from the location of the current research. Therefore, the study contributes to establishing the psychometric properties of these measures and their usefulness in the Nigerian context through reliability and validity. Finally, the result of this study confirmed the theories of social learning and social identity, as the data confirmed Hypotheses 1 and 2 based on their propositions.

6.5 Implications for practise

VPEB for employees is a highly sought-after employee action associated with desirable organisational consequences for the individual, the organisation, and society. This study found that EL and PO can independently enhance the expression of VPEB by employees. As EL and PO play crucial roles in achieving the VPEB of the employees, it is essential to introduce, improve, maximise, and maintain their presence within the organisation at all levels. These can be accomplished through strategic approaches such as employee recruitment, selection, training, and development. Other approaches to cultivating PO in an organisation include work design that gives employees a sense of autonomy and task identity. Employees' ability to be creative in their work allows them to participate in decisions that affect them, giving them access to information and resources and implementing formal ownership of shares or profits (Pickford et al., 2016; Tàpies & Lucía, 2013). Where the VPEB of employees is of concern, organisations must consider several characteristics associated with EL, such as equity, honesty, community, respect, integrity, responsibility, and accountability in selecting and promoting leaders. Similarly, organisations should implement employee participation in decision-making, profit-sharing institutes, and a sustained

sense of organisational ownership in employees. Some psychometrically good instruments are available to select, train and develop employees in EL and PO.

7 Conclusions

The VPEB of employees has been substantially related to organisational performance and sustainable development. These valuable results have required research efforts to provide an understanding that would guide the improvement of employee behaviour. The search for this concern primarily focussed on identifying the behaviour's plausible precursors. The current study investigated how EL and PO influence the VPEB of employees to support this concern. Importantly, this study revealed that EL and PO are precursors of the VPEB of employees.

Furthermore, EL has a more significant influence on VPEB than PO. However, the interaction of PO and EL does not significantly affect the VPEB of employees. These findings imply that EL and PO can be manipulated to improve the VPEB of employees. However, PO is a plausible option when a choice is necessary.

7.1 Limitations and recommendations for further research

When interpreting the results of this research, it is vital to consider its limitations, such as the adoption of a cross-sectional design and the reliance on self-reported data. These factors should be carefully considered before drawing any definitive conclusions from the study and formulating directions for future research. Self-reported data is subject to social desirability bias, whereas cross-sectional design does not allow observation and interpretation. Therefore, future research is encouraged to look for longitudinal, quasi-experimental, and multisource data to investigate the causation of the connections between VPEB of employees, EL, and PO. Second, this study did not cover the VPEB initiated by the organisation or part of the job requirement according to the adopted measure. Future research should examine the relative contribution of involuntary and VPEB. Third, the interaction model has EL as the sole predictor, PO as the sole moderator, and VPEB as the sole criterion variables. Despite treating certain demographic factors as covariates in the data analysis, it is essential to note that this study is a basic model and may still be susceptible to underspecification. Future studies should examine more complex models, such as moderated-mediated models. Fourth, to ensure the generalisability of the findings, this study sampled employees from public and private organisations. However, studies specific to ownership (public and private organisations) and activities (service and manufacturing organisations) are equally needed to build the literature. Finally, the observed interaction effect was non-significant, and the sample size could be implicated in this direction of the result as moderation effects are usually minor (Jex & Britt, 2014). Therefore, it is recommended that a larger sample be used in future studies.

Appendix 1

Sample Questionnaire for the Study.

Ethical Leadership

1. Listens to what employees have to say
2. Discipline employees who violate ethical standards.
3. Conduct your personal life ethically.
4. The best interest of the employees is in mind.
5. Make fair and balanced decisions.
6. Can be trusted
7. Discusses business ethics or values with employees
8. Set an example of how to do things correctly in terms of ethics.
9. Defines success not only by results but also by how they are obtained.
10. When making decisions, ask, "What is the right thing to do?"

Psychological ownership

1. This is my organisation.
2. I sense that this organisation is our company.
3. I feel a very high degree of personal ownership of this organisation.
4. I feel like this is my company.
5. This is our company.
6. Most people who work for this organisation feel like they own the company.
7. It is hard for me to think of an organisation like mine. (reversed)

Employees' voluntary pro-environmental behaviour.

1. I recycle at work when possible.
2. At work, I help my colleagues to be environmentally friendly.
3. At work, I make sure to conserve the number of materials that I use.
4. I encourage my colleagues to turn off work-related equipment when not in use.
5. At work, I promote environmentally friendly behaviour amongst my colleagues.
6. I persuade my manager to purchase environmentally friendly products for work purposes.
7. I always reduce the amount of energy I use at work.
8. I discussed with my manager how my organisation can become more environmentally friendly.
9. I encourage my organisation to support an environmental charity. (deleted)
10. I always encourage my organisation to reduce its environmental impact.

Acknowledgements The authors declared that the submitted manuscript is original research that has not been submitted or published elsewhere. To the authors' knowledge, every existing work referred to or used was appropriately acknowledged.

Funding Open access funding provided by Walter Sisulu University.

Data availability The data on which the results were based can be obtained on request from the second author.

Declarations

Competing interest The authors funded the study as no financial assistance was received from any organisation.

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